

Customer Support Routing Evaluation

Workbook analysis and two separate controlled development comparisons

This report keeps the assignment workbook and notebook experiments separate. The workbook-aligned prompt changes exact match from 145/150 to 146/150 with one improvement and zero regressions. The separate notebook prompt changes exact match from 91/100 to 98/100 with eight improvements and one regression. The datasets follow different multi-intent label policies, so their scores are never combined.

Evidence stream	Baseline	Focused	Case accounting	Decision
Workbook aligned	145/150	146/150	1 win, 0 regressions	Review: other precision 0.882
Notebook exercise	91/100	98/100	8 wins, 1 regression	Review: regression rate 1.10%

Evaluation objective

The router assigns one operational destination: order status, refund request, product issue, account help, or other. Incorrect routing can delay a monetary remedy, leave an account problem unresolved, or create an avoidable transfer. The evaluation therefore gives case-level regressions the same visibility as aggregate accuracy.

Required workbook analysis

The supplied workbook contains 150 tickets, 121 correct supplied predictions, and 29 supplied failures. Every failure received a human annotation and exactly one mechanism-based label. The four groups sum to all 29 failures.

Failure group	Cases	Observed risk
Refund Intent Underouted	12	Monetary-remedy requests can miss the correct workflow.
Account Journey Misrouted	9	Login, checkout, or account-record problems can remain unresolved.
Shipping Context Boundary	4	Shipping feedback and address incidents can reach the wrong route.
Product Context Boundary	4	Pre-purchase questions and received-product problems can be confused.

Step 2 · Group similar failures, give each group a title			
Re-read your Step 1 annotations. Cluster the failures that fail the SAME way. Give each cluster a short, memorable name. Aim for 2-4 clusters.			
Failure category title	One-line definition	Count	Why it matters (1 sentence)
Refund_Intent_Underouted	Explicit refund, money-back, price-adjustment, or cancellation-plus-refund intent is routed to another category.		12 Missing monetary intent delays resolution, creates transfers, and increases escalation risk for the largest failure group.
Account_Journey_Misrouted	Checkout, account visibility, or account-session failures are routed by nearby product or order words.		9 Customers can remain blocked from payment or purchase records when cases bypass the account-support workflow.
Shipping_Context_Boundary	The model confuses delivery-address incidents with profile changes, or generic shipping feedback with a live order.		4 These boundary errors send cases to teams with the wrong tools and service-level expectations.
Product_Context_Boundary	The model misses a post-purchase product-claim failure or treats a pre-purchase authenticity question as a defect.		4 Confusing product complaints with browsing questions causes avoidable returns work or leaves valid claims unresolved.

Workbook evidence: four human-defined groups covering all 29 supplied failures.

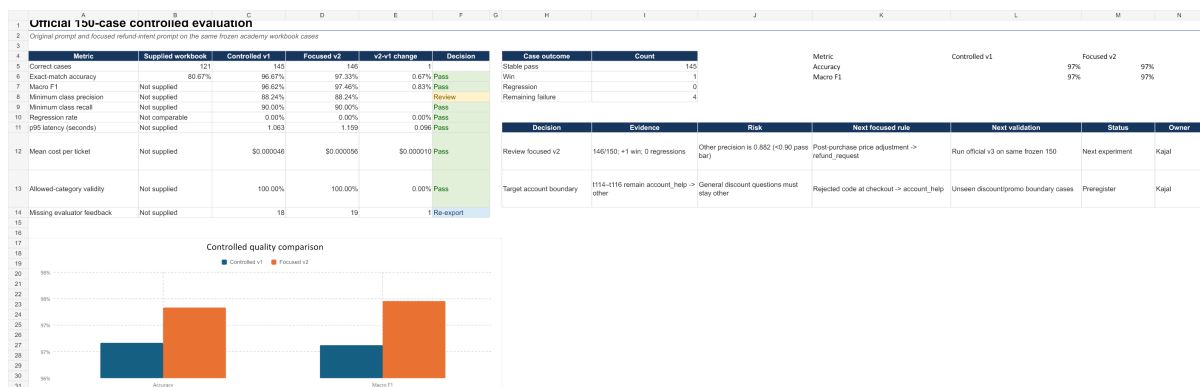
Workbook-aligned evaluation

Workbook Step 4 makes one focused change: it prioritizes explicit refund, money-back, price-adjustment, and cancellation-plus-refund intent. The fresh controlled original-prompt run scored 145/150. The Step 4 prompt scored 146/150 on the same rows and settings.

The supplied 121/150 snapshot and fresh 145/150 run are different artifacts. Their difference is model or runtime reproducibility evidence. Only the controlled 145-to-146 change is attributed to the focused prompt.

Outcome	Cases	Meaning
Stable pass	145	Correct under both controlled prompts
Wrong-to-right win	1	Incorrect under original prompt and correct under Step 4 prompt
Right-to-wrong regression	0	No original-prompt pass became incorrect
Remaining failure	4	Incorrect under both prompts

The intended refund group had little remaining headroom in the fresh baseline, and the only new win occurred in the shipping-context group. This is a modest net gain with no regressions. It does not prove the intended refund mechanism produced the improvement.



Workbook-aligned comparison: 145/150 to 146/150, one win, zero regressions, and four remaining failures.

The candidate remains Review because the `other` class precision is 0.882, below the 0.90 per-class bar. Deterministic output-versus-reference scoring covers all 150 rows even though some exported LangSmith feedback cells were blank.

Workbook operational tradeoff

Metric	Original prompt	Step 4 prompt	Change
Exact match	96.67%	97.33%	+0.67 points
Mean latency	0.740 s	0.771 s	+0.031 s
P95 latency	1.063 s	1.159 s	+0.096 s
Mean tokens	231.49	298.14	+66.65
Mean cost per case	\$0.000045623	\$0.000055911	+\$0.000010288
Total cost	\$0.00684345	\$0.00838665	+\$0.00154320

Workbook conclusion

The focused change produces a small improvement without a right-to-wrong regression. The per-class precision gate prevents promotion. The remaining cases show that the label policy also needs explicit treatment of post-purchase price adjustments and checkout-code failures, but those issues are reported as next hypotheses rather than silently repaired.

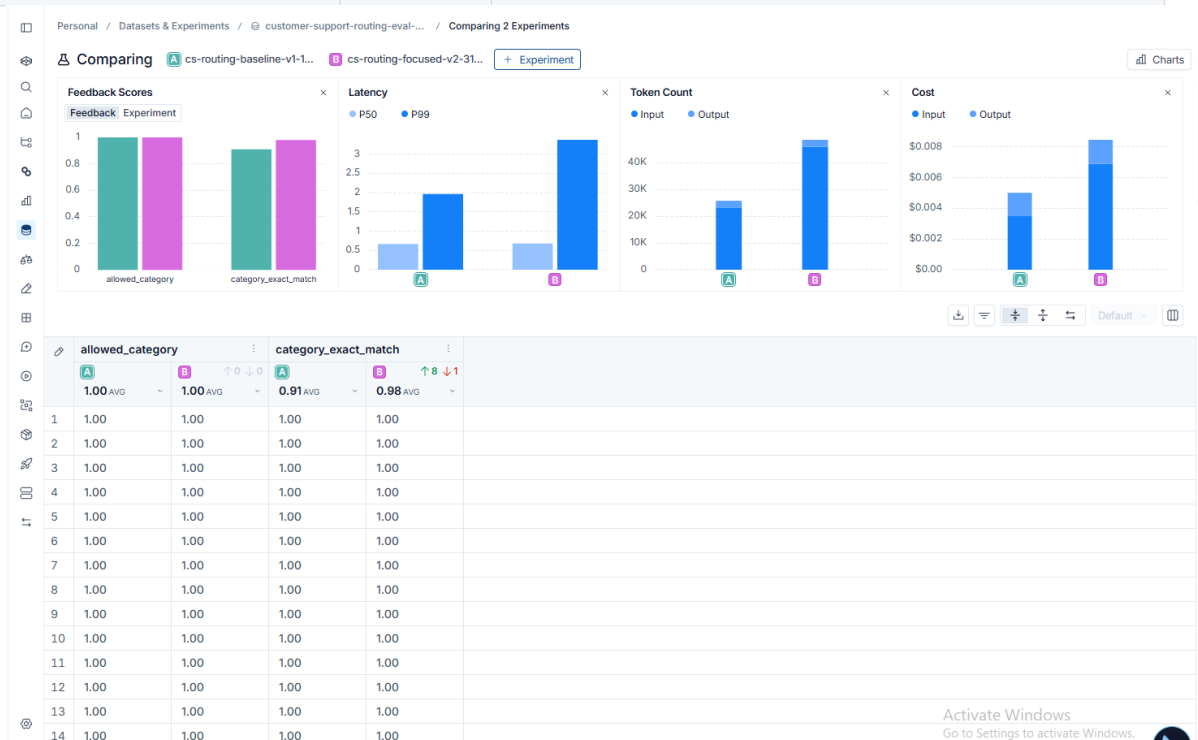
Evaluation methods

Method	Purpose
Exact match	Measures agreement with the reference category
Per-class precision, recall, and F1	Prevents overall accuracy from hiding weak routes
Allowed-category validity	Checks structured-output reliability
Wins and regressions	Shows who benefits and who is harmed by the change
Latency, tokens, and cost	Measures the operational tradeoff
Human failure review	Converts errors into mechanisms and an improvement hypothesis

Separate notebook evaluation

The executed notebook contains a separate 100-case development exercise under a different multi-intent routing policy. Its focused prompt emphasizes the underlying operational problem. This result supports notebook execution and LangSmith observability, but it is not presented as the execution of workbook Step 4.

Outcome	Cases	Meaning
Stable pass	90	Correct under both notebook prompts
Wrong-to-right win	8	Incorrect under notebook V1 and correct under notebook V2
Right-to-wrong regression	1	Correct under notebook V1 and incorrect under notebook V2
Stable failure	1	Incorrect under both notebook prompts



Notebook comparison: exact match 0.91 to 0.98 and allowed-category validity 1.00 for both prompts.

Four missing-order tickets move from refund request to order status, and four damaged-product tickets move from refund request to product issue. The regression and remaining failure concern misleading product-description language routed to other.

Notebook wins and regression

Personal / Datasets & Experiments / customer-support-routing-eval-... / Comparing 2 Experiments

Comparing A cs-routing-baseline-v1-1... B cs-routing-focused-v2-31... [+ Experiment](#)

Charts

Download Filter Sort Columns Default

	Inputs	Reference Outputs	Outputs		category_exact_match	
			A	B	A 0.91 AVG	B 0.98 AVG
1	Order #98765 is missing, and I'm pretty ... set about it. Can you just refund me alread	order_status	refund_request	order_status	0.00	1.00
2	The laptop I ordered came in with dama... and I'd prefer a full refund rather than a re	product_issue	refund_request	product_issue	0.00	1.00
3	My laptop arrived damaged and I want a ... ull refund, not a replacement.	product_issue	refund_request	product_issue	0.00	1.00
4	Hello, I'm reaching out because my orde... hasn't arrived yet. Could you please help	order_status	refund_request	order_status	0.00	1.00
5	I never received my order and I want my ... oney back.	order_status	refund_request	order_status	0.00	1.00
6	I'm really unhappy because my laptop sh... wed up broken, and I just want my money	product_issue	refund_request	product_issue	0.00	1.00
7	I've been waiting for my order, and it's st... not here! I demand a refund immediately!	order_status	refund_request	order_status	0.00	1.00
8	I received my laptop in a damaged condi... on and I expect a complete refund, not a r	product_issue	refund_request	product_issue	0.00	1.00

Eight notebook wrong-to-right improvements with reference labels and both outputs.

Personal / Datasets & Experiments / customer-support-routing-eval-... / Comparing 2 Experiments

Comparing A cs-routing-baseline-v1-1... B cs-routing-focused-v2-31... [+ Experiment](#)

Charts

Download Filter Sort Columns Default

	Inputs	Reference Outputs	Outputs		category_exact_match	
			A	B	A 0.91 AVG	B 0.98 AVG
1	The product description said 'wireless' b... I had to buy a separate dongle to use it. Mi	product_issue	product_issue	other	1.00	0.00

1 regression

The notebook right-to-wrong regression is shown separately with equal detail.

The notebook regression rate is 1 divided by 91 baseline-correct cases, or 1.10%. It narrowly exceeds the 1% regression guardrail. The 98% aggregate score therefore leads to Review rather than automatic promotion.

Notebook operational tradeoff

Metric	Baseline V1	Focused V2	Change
Mean latency	0.773 s	0.855 s	+0.082 s
P50 latency	0.678 s	0.678 s	No material change
P95 latency	1.028 s	1.510 s	+0.482 s
Mean tokens	241.52	476.46	+234.94
Mean cost per case	\$0.000046884	\$0.000082998	+\$0.000036114
Total cost	\$0.0046884	\$0.0082998	+\$0.0036114

LangSmith evidence

The notebook experiments contain 100 completed rows with exact-match and allowed-category feedback. The workbook-aligned experiments contain 150 completed outputs each, with deterministic reconciliation for every row. Preserved screenshots show aggregate comparisons, wins, the notebook regression, and representative traces.

Evidence	Artifact
Workbook comparison	14_official150_controlled_summary.png
Notebook aggregate	11a_langsmith_aggregate_scores.png
Notebook wins	11b_langsmith_eight_improvements.png
Notebook regression	11c_langsmith_one_regression.png
Trace evidence	09_corrected_trace_v2.png and 10_regression_trace_v2.png
Raw exports	Four versioned LangSmith CSV exports plus case reconciliations

Limitations

- Both comparisons use development data. Neither establishes organizer-hidden or production performance.
- The notebook and workbook follow different label policies for at least one multi-intent ticket. Their prompts, scores, and conclusions remain separate.
- The workbook-focused improvement is modest because the fresh original-prompt run already handled most cases in the targeted group.
- The notebook-focused prompt increases token use, cost, and p95 latency and introduces one regression.
- A stronger generalization claim requires an organizer-owned or independently authored untouched set frozen before the final prompt is scored.

Recommended next experiment

Freeze one final candidate before opening a new untouched validation source. The new source should follow one adjudicated label policy and be scored once. Any later prompt change requires a different untouched set. Retain exact match, per-class metrics, regressions, reliability, latency, token, and cost measures.

Submission evidence

The package includes the completed workbook, executed notebook, workbook-aligned and notebook LangSmith exports, case reconciliations, screenshots, an evidence index, and SHA-256 checksums. Every claim identifies its dataset and prompt pair.

Source material

- Week 4 Project Handout Aug 2026
- Customer Support Agent Evaluation Project Handout
- Evaluations Using LangSmith companion
- Week 4 AI Evals E Commerce Customer Support Agent workbook